
Homework 1

Michael Introini

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Chapter 2

Section 2.3

Problem 14

Determine whether $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is onto if:

b) $f(m, n) = m^2 - n^2$

ONTO

let $m^2 = 0$, then

$$-n^2 = y$$

$$n = \sqrt{y}$$

Then the pair is: $(0, \sqrt{y})$

$$f(0, \sqrt{y}) = 0 - \sqrt{y}^2$$

Suppose $y = 2$, then the function will not be onto.

c) $f(m, n) = m + n + 1$

NOT ONTO

let $n = -1$, this will allow m to fall in $\mathbb{Z}$

Problem 22

c) Determine whether each of these functions is a bijection from $\mathbb{R} \rightarrow \mathbb{R}$.

$$f(x) = (x+1)/(x+2)$$

Not a bijection from $\mathbb{R} \rightarrow \mathbb{R}$ because the $f(x)$ will never be equal to 1.

Problem 42

b) Let f be the function from $\mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$.

Find:

$$f^{-1}(\{x|0 < x < 1\})$$

$$f^{-1} = (-1, 0) \cup (0, 1)$$

Problem 44

a) Let f be a function from A to B . Let S and T be subsets of B . Show that:

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$$

Section 2.4

Problem 30b

What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

$$\mathbf{b) \sum_{j \in S} j^2 = 1 + 9 + 25 + 49 = 84}$$

Problem 34b

Compute each of these double sums.

$$\begin{aligned} \mathbf{b) \sum_{i=0}^3 \sum_{j=0}^2 (3i+2j)} \\ = \sum_{i=0}^3 i = 0^2 = (3i+0+3i+2+3i+4) \\ = \sum_{i=0}^3 i = 0^3 = (9i+6) = ((9x0+6) + (9x1+6) + (9x2+6) + (9x3+6)) \\ = 15 + 24 + 33 = \mathbf{78} \end{aligned}$$

Problem 36

Use the identity $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ and Exercise 35 to compute $\sum_{k=1}^n \frac{1}{k(k+1)}$.

$$\begin{aligned}\sum_{k=1}^n \frac{1}{k(k+1)} &= \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^n \frac{1}{k+1} \\ \sum_{k=1}^n \frac{1}{k} - \frac{1}{k+1} &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \left(-\frac{1}{4} + \frac{1}{4}\right) + \cdots + \left(-\frac{1}{n} + \frac{1}{n}\right) - \frac{1}{n+1} = 1 - \frac{1}{n+1} \\ &= \frac{n}{n+1}\end{aligned}$$

Problem 44

Express $n!$ using product notation

$$n! = n(n-1)(n-2)\dots(n-(n-1))$$

Section 2.5

Problem 4c,d

Determine whether each of these sets is countable or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.

c) the real numbers with decimal representations consisting of all 1s.

Countable

To prove that these numbers are countable, we can construct an array of ordered pairs of $\mathbb{N} \times \mathbb{N}$ for which we can make a one to one correspondence out of.

(1, 1)	(2, 1)	(3, 1)
(1, 2)	(2, 2)	(3, 2)
(1, 3)	(2, 3)	(3, 3)

d) the real numbers with decimal representations of all 1s or 9s.

Uncountable

To prove this is uncountable, we can use Cantor's Diagonalization Method:
Suppose we have:

$r_1 = 1.119911$
 $r_2 = 1.199119$
 $r_2 = 1.919111$

We construct a new number using the following rule:

$$d_i = \begin{cases} 1 & \text{if } d_{ii} \neq 1 \\ 9 & \text{if } d_{ii} = 1 \end{cases}$$

We can now prove that this set is uncountable.

Section 2.6

Problem 4c

Evaluate AB

$$A = \begin{bmatrix} 0 & -1 \\ 7 & 2 \\ -4 & -3 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & -1 & 2 & 3 & 0 \\ -2 & 0 & 3 & 4 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 0 & -3 & -4 & -1 \\ 24 & -7 & 20 & 29 & 2 \\ -10 & 4 & -17 & -24 & -3 \end{bmatrix}$$

Chapter 3

3.1

Problem 54a,b

Use the greedy algorithm to make change using quarters, dimes, and pennies (but no nickels) for each of the amounts given in Exercise 52. For which of these amounts does the greedy algorithm use the fewest coins of these denominations possible?

	Change	Coin Denomination
a)	87	1 Quarter
	62	1 Quarter
	37	1 Quarter
	12	1 Dime
	2	1 Penny
	1	1 Penny
	Total Coins	6 Coins

This is the smallest amount of coins possible for this algorithm.

Change	Coin Denomination
49	1 Quarter
24	1 Dime
14	1 Dime
b) 4	1 Penny
3	1 Penny
2	1 Penny
1	1 Penny
Total Coins	7 Coins

This is the smallest amount of coins possible for this algorithm.

Problem 56

Show that if there were a coin worth 12 cents, the greedy algorithm using quarters, 12-cent coins, dimes, nickels, and pennies would not always produce change using the fewest coins possible.

Suppose we had to return 15 cents, then we would get the following outcome:

Change	Coin Denomination
15	12 Cent Coin
3	1 Penny
2	1 Penny
1	1 Penny
Total Coins	4 Coins

Instead it would be more efficient without the 12 cent coin since we could return 1 dime and 1 nickel for a total of 2 coins.

Section 3.2

Problem 8

Find the least integer n such that $f(x)$ is $O(x^n)$ for each of these functions.

a) $f(x) = 2x^2 + x^3 \log x$

b) $f(x) = 3x^5 + (\log x)^4$

c) $f(x) = (x^4 + x^2 + 1)/(x^4 + 1)$

d) $f(x) = (x^3 + 5 \log x)/(x^4 + 1)$

a)

Simplify to $f(x) = O(x^3 \log x)$

We now want $x^3 \log x = O(x^n)$

Since x^3 is still $< \log x$ we can use x^4 as the least integer n to make this $O(x^n)$

b)

The term x^5 dominates the function so this is already at $O(x^n)$

c)

As x^4 grows it gets closer to 1, which then can be written as $O(1)$, this can also be written as $O(x^0)$.

d)

Simplify to $f(x) = O(\frac{1}{x^4})$

Since we want $O(x^n)$ we can use the power $(\frac{1}{x^4})^{-1}$ to invert and turn this into $O(x^4)$

Problem 26

Give a big-O estimate for each of these functions. For the function g in your estimate $f(x)$ is $O(g(x))$, use a simple function g of smallest order.

a) $(n^3 + n^2 \log n)(\log n + 1) + (17 \log n + 19)(n^3 + 2)$

Taking the largest terms we get: $= n^3 \log n + n^3 \log n$

By Theorem 2 we can say that the entire function is $O(n^3 \log n)$

b) $(2^n + n^2)(n^3 + 3^n)$

By Theorem 3 this becomes $O(2^n 3^n)$

c) $(n^n + n2^n + 5^n)(n! + 5^n)$

By Theorem 3 this becomes $O(n! n^n)$

Problem 38

Give a big-O estimate of the product of the first n odd positive integers.

$O((2n)^n)$

Chapter 4

Section 4.1

Problem 20c

Evaluate these quantities.

c) $-101 \bmod 13$

3

Section 4.2

Problem 2a

Convert the decimal expansion of each of these integers to a binary expansion.

a 321

$(101000001)_2$

Problem 4c

Convert the binary expansion of each of these integers to a decimal expansion.

c) $(1110111110)_2$

$$1 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 958$$

Section 4.3

Problem 2f

Determine whether each of these integers is prime.

f 113

113 is Prime, there exists no other integer that factors into it.

Problem 4d

Find the prime factorization of each of these integers

d 143

$11 \cdot 13$

Problem 6

Problem 12

Problem 18a

Problem 24b

Problem 26b

Problem 30

Problem 32c

Section 4.4

Problem 4

Problem 10